PRISMA 2020 Eligibility Stage: Quality Criteria Scoring

Report prepared 30 August 2026

1. Input to the Eligibility Step

Step 4 of the eligibility stage was run on the 141 records that survived Step 3 (title and abstract screening against the inclusion and exclusion criteria). Each of these 141 full texts was assessed against the quality criteria set out in Section 2 to determine eligibility for inclusion in the synthesis stage.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each.

Table 1. Quality assessment criteria applied in the Step 4 eligibility run

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

A record was retained for the next step, namely synthesis, only where its quality score met or exceeded 2.5 out of a possible 4, or the equivalent proportion where QC4 was not applicable. Venue indexing alone could not satisfy this threshold, since QC3 contributes at most one of the four points, so a study additionally had to demonstrate at least partial strength across problem clarity, methodological transparency, or empirical detail to be retained. For IC2 studies, for which QC4 does not apply, the same proportional standard corresponds to a reduced threshold of 1.5 out of a possible 3.

3. Results

Of the 141 records assessed against the quality criteria in Section 2, 87 records met or exceeded their applicable threshold and were retained for synthesis, while 54 records fell below their applicable threshold and were excluded at this step. Table 2 lists each assessed record together with its total quality score, applicable threshold, and eligibility decision, sorted with included records listed first, followed by excluded records.

Table 2. Quality criteria scoring results, sorted by decision

No.	Title (Year)	Section	Total Score	Threshold	Decision
1	A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security [2020]	1.0	4	>= 2.5	**Include**

No.	Title (Year)	Section	Total Score	Threshold	Decision
2	Machine Learning in IoT Security: Current Solutions and Future Challenges [2020]	1.0	4	>= 2.5	**Include**
3	Concept Drift Under Harsh Constraints: A Review of Potential Strategies for IoT Systems [2025]	1.0	4	>= 2.5	**Include**
4	A Taxonomy of Machine-Learning-Based Intrusion Detection Systems for the Internet of Things: A Survey [2022]	1.0	4	>= 2.5	**Include**
5	A Survey of Deep Learning Technologies for Intrusion Detection in Internet of Things [2024]	1.0	4	>= 2.5	**Include**
6	A comprehensive survey on deep learning-based intrusion detection systems in Internet of Things (IoT) [2025]	1.0	3	>= 2.5	**Include**
7	Recent endeavors in machine learning-powered intrusion detection systems for the Internet of Things [2024]	1.0	3.5	>= 2.5	**Include**
8	A Comprehensive Survey on Concept-Drift-Resilient Network Intrusion Detection Systems [2026]	1.0	4	>= 2.5	**Include**
9	Network Intrusion Detection for IoT Security Based on Learning Techniques [2019]	1.0	4	>= 1.5	**Include**
10	Internet of Things: A survey on machine learning-based intrusion detection approaches [2019]	1.0	2.5	>= 1.5	**Include**
11	A Survey on Transfer Learning [2010]	3.1	4	>= 1.5	**Include**
12	Improving predictive inference under covariate shift by weighting the log-likelihood function [2000]	3.1	4	>= 1.5	**Include**
13	A unifying view on dataset shift in classification [2012]	3.1	4	>= 1.5	**Include**
14	A unified framework for dataset shift diagnostics [2023]	3.1	4	>= 2.5	**Include**
15	Efficient and Multiply Robust Risk Estimation under General Forms of Dataset Shift [2024]	3.1	4	>= 2.5	**Include**
16	A Survey of Unsupervised Deep Domain Adaptation [2020]	3.1	3.5	>= 2.5	**Include**
17	A Comprehensive Survey on Test-Time Adaptation Under Distribution Shifts [2025]	3.1	3.5	>= 2.5	**Include**
18	Evaluation of domain generalization and adaptation on improving model robustness to temporal dataset shift in clinical medicine [2022]	3.1	4	>= 2.5	**Include**

No.	Title (Year)	Section	Total Score	Threshold	Decision
19	When Invariant Representation Learning Meets Label Shift: Insufficiency and Theoretical Insights [2024]	3.1	4	>= 2.5	**Include**
20	Preventing dataset shift from breaking machine-learning biomarkers [2021]	3.1	3	>= 2.5	**Include**
21	Handling Out-of-Distribution Data: A Survey [2025]	3.1	4	>= 2.5	**Include**
22	A Survey on Concept Drift Adaptation [2014]	3.2	4	≥1.5	**Include**
23	Learning under Concept Drift: A Review [2019]	3.2	4	≥1.5	**Include**
24	Characterizing Concept Drift [2016]	3.2	4	≥1.5	**Include**
25	A Survey of Processing Methods for Different Types of Concept Drift [2026]	3.2	4	≥2.5	**Include**
26	From Concept Drift to Model Degradation: An Overview on Performance-Aware Drift Detectors [2022]	3.2	4	≥2.5	**Include**
27	One or Two Things We Know About Concept Drift — A Survey on Monitoring in Evolving Environments. Part A: Detecting Concept Drift [2024]	3.2	4	≥2.5	**Include**
28	Table for estimating the goodness of fit of empirical distributions [1948]	3.3	4	>= 1.5	**Include**
29	Learning with Drift Detection [2004]	3.3	4	>= 1.5	**Include**
30	Reactive Soft Prototype Computing for Concept Drift Streams [2020]	3.3	4	>= 2.5	**Include**
31	Detection of data drift in a two-dimensional stream using the Kolmogorov-Smirnov test [2022]	3.3	4	>= 2.5	**Include**
32	Diversity measure as a new drift detection method in data streaming [2020]	3.3	4	>= 2.5	**Include**
33	Online kernel CUSUM for change-point detection [2026]	3.3	4	>= 2.5	**Include**
34	Multiple Comparisons among Means [1961]	3.3	4	>= 1.5	**Include**
35	Drift adaptive online DDoS attack detection framework for IoT system [2024]	4.1	4	>= 2.5	**Include**
36	A Network Intrusion Detection System for Concept Drifting Network Traffic Data [2021]	4.1	4	>= 2.5	**Include**
37	CDDA-MD: An efficient malicious traffic detection method based on concept drift detection and adaptation technique [2025]	4.1	4	>= 2.5	**Include**

No.	Title (Year)	Section	Total Score	Threshold	Decision
38	Intrusion detection in the IoT data streams using concept drift localization [2024]	4.1	4	>= 2.5	**Include**
39	Concept Drift Analysis Based on Isolation Forest for Effectively Detecting Network Attack in IoT Scenarios [2026]	4.1	4	>= 2.5	**Include**
40	SFC-NIDS: a sustainable and explainable flow filtering based concept drift-driven security approach for network introspection [2024]	4.1	4	>= 2.5	**Include**
41	Class imbalance and concept drift invariant online botnet threat detection framework for heterogeneous IoT edge [2024]	4.1	4	>= 2.5	**Include**
42	Self-Supervised Adaptation Method to Concept Drift for Network Intrusion Detection [2025]	4.1	4	>= 2.5	**Include**
43	Enhanced Intrusion Detection with Data Stream Classification and Concept Drift Guided by the Incremental Learning Genetic Programming Combiner [2023]	4.1	4	>= 2.5	**Include**
44	Feature Drift Aware for Intrusion Detection System using Developed Variable Length Particle Swarm Optimization in Data Stream [2023]	4.2	4	>= 2.5	**Include**
45	Intrusion Detection in the IoT under Data and Concept Drifts: Online Deep Learning Approach [2022]	4.2	4	>= 2.5	**Include**
46	Unsupervised Incremental Learning with Dual Concept Drift Detection for Identifying Anomalous Sequences [2024]	4.2	3.5	>= 2.5	**Include**
47	A Representation Learning Approach to Feature Drift Detection in Wireless Networks [2026]	4.2	3.5	>= 2.5	**Include**
48	Addressing Concept Drift in IoT Anomaly Detection: Drift Detection, Interpretation, and Adaptation [2024]	4.2	4	>= 2.5	**Include**
49	Reinforcement Learning-Based Voting for Feature Drift-Aware Intrusion Detection: An Incremental Learning Framework [2025]	4.2	4	>= 2.5	**Include**
50	From one attack domain to another: Contrastive transfer learning with siamese networks for APT detection [2026]	4.2	3.5	>= 2.5	**Include**
51	MOA: Massive Online Analysis [2010]	4.3	4	≥ 1.5	**Include**

No.	Title (Year)	Section	Total Score	Threshold	Decision
52	Stream-learn - open-source Python library for difficult data stream batch analysis [2022]	4.3	4	≥ 2.5	**Include**
53	Challenges in benchmarking stream learning algorithms with real-world data [2020]	4.3	4	≥ 2.5	**Include**
54	Synthetic non-stationary data streams for recognition of the unknown [2026]	4.3	4	≥ 2.5	**Include**
55	Concept Drift Detection in Data Stream Mining: A literature review [2022]	4.3	4	≥ 2.5	**Include**
56	A Survey of Data Stream-Based Intrusion Detection Systems [2025]	5.1	4	>= 2.5	**Include**
57	A comparative study on online machine learning techniques for network traffic streams analysis [2022]	5.1	4	>= 2.5	**Include**
58	Evolving Fuzzy-Rule-Based Classifiers From Data Streams [2008]	5.1	3	>= 1.5	**Include**
59	Incremental on-line learning: A review and comparison of state of the art algorithms [2018]	5.1	4	>= 1.5	**Include**
60	An Incremental Learning Method Based on Dynamic Ensemble RVM for Intrusion Detection [2022]	5.1	4	>= 2.5	**Include**
61	IL-IDS: An incremental learning approach with confined data streams for intrusion detection [2025]	5.1	4	>= 2.5	**Include**
62	Concept Drift-Based Intrusion Detection For Evolving Data Stream Classification In IDS: Approaches And Comparative Study [2024]	5.1	4	>= 2.5	**Include**
63	The Impact of Diversity on Online Ensemble Learning in the Presence of Concept Drift [2010]	5.1	4	>= 1.5	**Include**
64	Three types of incremental learning [2022]	5.2	4	>= 2.5	**Include**
65	Overcoming catastrophic forgetting in neural networks [2017]	5.2	4	>= 1.5	**Include**
66	Continual lifelong learning with neural networks: A review [2019]	5.2	4	>= 1.5	**Include**
67	Learning without Forgetting [2018]	5.2	3.5	>= 1.5	**Include**
68	Continual Learning With Knowledge Distillation: A Survey [2025]	5.2	3.5	>= 2.5	**Include**
69	Overcoming Class Imbalance in Incremental Learning Using an Elastic Weight Consolidation-Assisted Common Encoder Approach [2025]	5.2	4	>= 2.5	**Include**

No.	Title (Year)	Section	Total Score	Threshold	Decision
70	XDFC-IDS: An explainable decentralized federated class-incremental fusion framework for intrusion detection [2026]	5.2, 5.3	4	>= 2.5	**Include**
71	Addressing catastrophic forgetting in class-incremental learning-A survey [2026]	5.2	3.5	>= 2.5	**Include**
72	A continual learning survey: Defying forgetting in classification tasks [2022]	5.2	3.5	>= 2.5	**Include**
73	Advances and open problems in federated learning [2021]	5.3	3.5	>= 2.5	**Include**
74	Federated continual learning: A comprehensive survey on lifelong and privacy-preserving learning over distributed and non-stationary data [2026]	5.3	4	>= 2.5	**Include**
75	Non-IID data and continual learning processes in federated learning: A long road ahead [2022]	5.3	4	>= 2.5	**Include**
76	FD-IDS: Federated learning with knowledge distillation for intrusion detection in Non-IID IoT environments [2025]	5.3	4	>= 2.5	**Include**
77	FL-MalDrift: A federated learning framework for malware detection under local concept drift [2026]	5.3	4	>= 2.5	**Include**
78	ATTA-FL-Lite: Lightweight byzantine-robust federated learning for resource-constrained medical IoT devices [2026]	5.3	4	>= 2.5	**Include**
79	Secure and explainable federated learning for IoT intrusion detection: A comprehensive survey [2026]	5.3	4	>= 2.5	**Include**
80	Explainable cross-domain evaluation of ML-based network intrusion detection systems [2023]	6.0	4	>= 2.5	**Include**
81	Inter-dataset generalization strength of supervised machine learning methods for intrusion detection [2020]	6.0	4	>= 2.5	**Include**
82	Benchmarking and Cross-Dataset Evaluation of AI-Based Intrusion Detection Systems for Smart City IoT Networks [2026]	6.0	4	>= 2.5	**Include**
83	Evaluating the Generalization Gaps of Intrusion Detection Systems Across DoS Attack Variants [2025]	6.0	4	>= 2.5	**Include**
84	Concept Drift Analysis by Dynamic Residual Projection for Effectively Detecting Botnet Cyber-Attacks in IoT Scenarios [2022]	6.0	4	>= 2.5	**Include**

No.	Title (Year)	Section	Total Score	Threshold	Decision
85	FORT-IDS: A federated, optimized, robust and trustworthy intrusion detection system for IIoT security [2025]	6.0	3.5	>= 2.5	**Include**
86	Dataset-centric evaluation of federated intrusion detection models in IoT networks [2026]	6.0	4	>= 2.5	**Include**
87	DI-NIDS: Domain invariant network intrusion detection system [2023]	6.0	4	>= 2.5	**Include**
88	Machine learning based solutions for security of Internet of Things (IoT): A survey [2020]	1.0	1.5	>= 2.5	**Exclude**
89	Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey [2023]	1.0	2	>= 2.5	**Exclude**
90	A critical review of intrusion detection systems in the internet of things: techniques, deployment strategy, validation strategy, attacks, public datasets and challenges [2021]	1.0	2	>= 2.5	**Exclude**
91	A Review on Machine Learning and Deep Learning Perspectives of IDS for IoT: Recent Updates, Security Issues, and Challenges [2020]	1.0	1.5	>= 2.5	**Exclude**
92	Network intrusion datasets: A survey, limitations, and recommendations [2025]	1.0	2.0	>= 2.5	**Exclude**
93	Deep learning for intrusion detection in emerging technologies: a comprehensive survey and new perspectives [2025]	1.0	2	>= 2.5	**Exclude**
94	Machine Learning-Enabled IoT Security: Open Issues and Challenges Under Advanced Persistent Threats [2022]	1.0	1.5	>= 2.5	**Exclude**
95	On the value of label and semantic information in domain generalization [2023]	3.1	2	>= 2.5	**Exclude**
96	Towards Accurate and Robust Domain Adaptation Under Multiple Noisy Environments [2023]	3.1	2	>= 2.5	**Exclude**
97	Generalization Error Bounds for Graph Neural Networks in Unseen Domains [2025]	3.1	2	>= 2.5	**Exclude**
98	Ensemble Approaches for Dynamic Data Stream Classification Under Label Scarcity [2025]	3.2	2	≥2.5	**Exclude**
99	Adaptation Strategies for Automated Machine Learning on Evolving Data [2021]	3.2	2	≥2.5	**Exclude**

No.	Title (Year)	Section	Total Score	Threshold	Decision
100	Type-LDD: A Type-Driven Lite Concept Drift Detector for Data Streams [2024]	3.2	2	≥2.5	**Exclude**
101	Concept Drift Adaptation by Exploiting Drift Type [2024]	3.2	2	≥2.5	**Exclude**
102	Dynamical Targeted Ensemble Learning for Streaming Data With Concept Drift [2024]	3.2	2	≥2.5	**Exclude**
103	An Integrated Preprocessing and Drift Detection Approach With Adaptive Windowing for Fraud Detection in Payment Systems [2025]	3.3	2	>= 2.5	**Exclude**
104	Novel hybrid pair recommendations based on a large-scale comparative study of concept drift detection [2021]	3.3	2	>= 2.5	**Exclude**
105	Bayesian Nonparametric Unsupervised Concept Drift Detection for Data Stream Mining [2020]	3.3	2	>= 2.5	**Exclude**
106	Adaptive telecom churn prediction for concept-sensitive imbalance data streams [2022]	3.3	2	>= 2.5	**Exclude**
107	N-DD: New Approach for Drift Detection Based on Neutrosophic Support Vector Machine [2025]	3.3	1	>= 2.5	**Exclude**
108	VLSA-CL: A variational latent space alignment and contrastive learning framework for robust detection of IoT botnet attacks under concept drift [2026]	4.1	2	>= 2.5	**Exclude**
109	FADES: Adaptive Drift Estimation via Conformal Signals for Streaming Intrusion Detection [2026]	4.1	2	>= 2.5	**Exclude**
110	DFAID: Density-aware and feature-deviated active intrusion detection over network traffic streams [2022]	4.1	2	>= 2.5	**Exclude**
111	Fast & Furious: On the modelling of malware detection as an evolving data stream [2023]	4.1	2	>= 2.5	**Exclude**
112	ADDAEIL: Anomaly Detection with Drift-Aware Ensemble-Based Incremental Learning [2025]	4.1	2	>= 2.5	**Exclude**
113	Real-Time Intrusion Detection Using Adaptive Sliding Windows for Concept Drift [2025]	4.1	1.5	>= 2.5	**Exclude**
114	Anomalies Detection Using Isolation in Concept-Drifting Data Streams [2021]	4.1	2	>= 2.5	**Exclude**
115	DyP-CNX: A Dynamic Preprocessing-Enhanced Hybrid	4.2	2	>= 2.5	**Exclude**

No.	Title (Year)	Section	Total Score	Threshold	Decision
	Model for Network Intrusion Detection [2025]				
116	A Framework for Drift Detection and Adaptation in AI-driven Anomaly and Threat Detection Systems [2025]	4.2	2	>= 2.5	**Exclude**
117	A Novel Method for Drift Detection in Streaming Data Based on Measurement of Changes in Feature Ranks [2025]	4.2	2	>= 2.5	**Exclude**
118	Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems [2024]	4.2	2	>= 2.5	**Exclude**
119	DRIFTNET-EnVACK: Adaptive Drift Detection in Cloud Data Streams With Ensemble Variational Auto-Encoder Featuring Contextual Network [2024]	4.3	2	≥ 2.5	**Exclude**
120	Concept drift detection methods based on different weighting strategies [2024]	4.3	2	≥ 2.5	**Exclude**
121	Concept anchor drift detection: a framework for proactive detection of abrupt drifts in real-world data streams [2026]	4.3	2	≥ 2.5	**Exclude**
122	Concept drift detection in district heating: A characterization framework and detector-centric benchmark [2026]	4.3	2	≥ 2.5	**Exclude**
123	Unsupervised Feature Selection for Outlier Detection on Streaming Data to Enhance Network Security [2021]	5.1	2	>= 2.5	**Exclude**
124	An online ensemble classification algorithm for multi-class imbalanced data stream [2024]	5.1	2	>= 2.5	**Exclude**
125	Active Weighted Aging Ensemble for drifted data stream classification [2023]	5.1	2	>= 2.5	**Exclude**
126	A Four-stage Bayesian Incremental Learning Framework for Intrusion Detection in Real-world IoT Environment [2026]	5.1	2	>= 2.5	**Exclude**
127	Towards anti-forgetting with masked optimal transport regularization for continual named entity recognition [2026]	5.2	2	>= 2.5	**Exclude**
128	Variance-Covariance Regularization Improves Continual Learning [2025]	5.2	2	>= 2.5	**Exclude**
129	Class Incremental Learning with probability dampening and cascaded gated classifier [2025]	5.2	2	>= 2.5	**Exclude**

No.	Title (Year)	Section	Total Score	Threshold	Decision
130	Class-Incremental Learning of Convolutional Neural Networks Based on Double Consolidation Mechanism [2020]	5.2	1.5	>= 2.5	**Exclude**
131	Federated learning in intrusion detection: advancements, applications, and future directions [2025]	5.3	2	>= 2.5	**Exclude**
132	Fed-CALiBER: Federated Lightweight BERT Intrusion Detection on CAN Bus Protocol in Autonomous Vehicle [2025]	5.3	2	>= 2.5	**Exclude**
133	Federated continual learning with domain-routed historical expert matching for multimodal perception [2026]	5.3	2	>= 2.5	**Exclude**
134	Adaptive Hierarchical Federated Learning for IoT Anomaly Detection [2025]	5.3	2	>= 2.5	**Exclude**
135	Non-IID and aware federated intrusion detection with PBFT with secured model aggregation for multi institutional healthcare internet of things networks [2026]	5.3	2	>= 2.5	**Exclude**
136	Communication-Efficient Federated Class-Incremental Intrusion Detection for Edge IoT Networks [2026]	5.3	2	>= 2.5	**Exclude**
137	SHAP-Guided Feature Selection for Cross-Dataset Generalization in Network Intrusion Detection Systems [2026]	6.0	2	>= 2.5	**Exclude**
138	Machine Learning in Network Intrusion Detection: A Cross-Dataset Generalization Study [2024]	6.0	2	>= 2.5	**Exclude**
139	Toward Generalization and Interpretable Deep Learning-Based Intrusion Detection System for Heterogeneous Network Environments [2026]	6.0	2	>= 2.5	**Exclude**
140	A Decentralized Federated WGAN-GP Framework for Anomaly Detection in Industrial Control Systems [2026]	6.0	2	>= 2.5	**Exclude**
141	A Systematic Review of Intrusion Detection Systems for Internet of Medical Things: Performance, Efficiency, Explainability, and Generalization [2026]	6.0	2	>= 2.5	**Exclude**

This retained set constitutes the final included-studies pool for this systematic literature review and forms the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each record was scored against the four quality criteria on the basis of its full text, and the retain or exclude decision was applied strictly according to the threshold in effect for that record, namely 2.5 out of 4 for IC1-captured records, or 1.5 out of 3 for IC2-captured records for which QC4 did not apply. No record was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Verification

This report has been prepared, checked, and approved in accordance with the review team's internal quality assurance procedure prior to proceeding to the synthesis stage.

Prepared by	Checked by	Approved by
Name: RAZIMAN ZAKARIA	Name: NOOR HAYATI MOHD ZAIN	Name: MOHD MURTADHA MOHAMAD